

# Structured Iterative Approximations in Numerical Linear Algebra

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# Introduction

# The "What?", the "Why?" and the "How?"

- A few *approximation problems* of interest to us:

- **Linear solves** with  $A \in \mathbb{R}^{n \times n}$  and  $b \in \mathbb{R}^n$ :

$$\text{Find } x \in \mathbb{R}^n \text{ s.t. } Ax \approx b.$$

- **Inversion** of  $A \in \mathbb{R}^{n \times n}$ :

$$\text{Find } M \in \mathbb{R}^{n \times n} \text{ s.t. } M \approx A^{-1}.$$

- **Low-rank CP representation** of  $\mathfrak{X} \in \mathbb{R}^{n_1 \times \dots \times n_d}$ :

$$\text{Find } (U_1, \dots, U_d) \in \mathbb{R}^{n_1 \times r} \times \dots \times \mathbb{R}^{n_d \times r} \text{ s.t. } \llbracket U_1, \dots, U_d \rrbracket \approx \mathfrak{X}$$

for some  $r \ll \min\{n_1, \dots, n_d\}$  where

$$\llbracket U_1, \dots, U_d \rrbracket := \sum_{k=1}^r \underbrace{U_1[:, k]}_{u_k^{(1)} \in \mathbb{R}^{n_1}} \circ \dots \circ \underbrace{U_d[:, k]}_{u_k^{(d)} \in \mathbb{R}^{n_d}}.$$

- Interests in both:

- unconstrained variants of these problems, and the case of
- **structured** (e.g., sparse, nonnegative, symmetric, definite, ...) iterates.

# Sparse Approximate Inverses (SPAIs)

## Sparse approximate inverses

- ▶ Numerical linear algebraic (NLA) problem of interest:
  - Finding an approximate inverse of a matrix  $A \in \mathbb{R}^{n \times n}$ :

$$\boxed{M \in \mathbb{R}^{n \times n} \text{ s.t. } AM \approx I_n}$$

such that  $x \mapsto Mx$  can be computed efficiently irrespective of  $x$ .

- ▶ Approximate inverses  $M$ , or at least, maps of the form  $x \mapsto Mx$ , are routinely used in NLA to accelerate the convergence of iterative solvers. For example,

Instead of solving for  $x$  s.t.  $Ax = b$ ,

we solve for  $u$  s.t.  $AMu = b$ .

Then, we have  $x = Mu$ .

- ▶ When  $A$  is sparse (i.e., most components are zero), although  $A^{-1}$  is generally dense (i.e., not sparse), sparse approximate inverse (SPAI) matrices  $M$  can nevertheless be good candidates to accelerate iterative computations by preconditioning.

## Best SPAs

- ▶ For prescribed sparsity patterns (i.e., for a given set of nonzero components in  $M$ ), we can leverage the fact that

$$\|I_n - AM\|_F^2 = \sum_{i=1}^n \|e_i^{(n)} - Am_i\|_2^2 \text{ where } m_i := Me_i^{(n)}$$

(the  $i$ -th column of  $M$ )

to find the "best" SPAI  $M$  through decoupled sparse least squares solves.

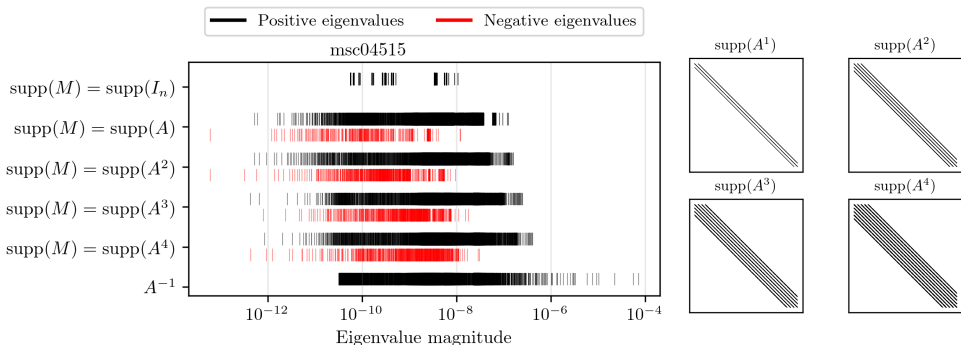
- ▶ We generally do not know what the sparsity pattern of  $M$  should be to efficiently accelerate iterative computations with  $A$ .

Standard choices are nonzero patterns based on monomials of  $A$ , i.e., sparsity patterns of  $A$ , of  $A^2$ , of  $A^3$ , ...

But those choices sometimes fail to yield good SPAs.

# Limits of best SPAs with sparsity of low powers of $A$

- ▶ Example of matrix for which the best SPAI with sparsity patterns of  $A, A^2, A^3, \dots$  fail to reproduce the definiteness of  $A^{-1}$ :



Alternative strategies exist that iterate from one sparsity pattern to another, gradually building a better SPAI.

We call those approaches sparse matrix iterations.

## State-of-the-art sparse matrix iteration

- ▶ Starting with an initial SPAI  $M_0$ , sparse matrix iterations build sequences of SPAI iterates  $M_1, M_2, \dots$  by trying to minimize

$$f(M) := \|I_n - AM\|_F^2 \quad \text{with} \quad \nabla_M f(M) \propto A^T R$$

where  $R := I_n - AM$  denotes the residual.

- ▶ The Newton-Schulz iteration is given by:

$$M_{i+1} := M_i(3I_n - M_i^2)/2.$$

- ▶ Steepest descent (SD) iterations are defined by the recurrence formula

$$M_{i+1} := \arg \min_{M \in M_i + \text{span}\{A^T R_i\}} f(M)$$

which is equivalently defined by the following projection:

$$M_{i+1} \in M_i + \text{span}\{A^T R_i\} \text{ s.t. } R_{i+1} := I_n - AM_{i+1} \perp A \text{span}\{A^T R_i\}.$$

## State-of-the-art sparse matrix iteration, cont'd

- ▶ Faster convergence than Newton-Schulz and SD is achieved by minimal residual (MR) iteration:

$$M_{i+1} := \arg \min_{M \in M_i + \text{span}\{R_i\}} f(M)$$

which is achieved by the following projection:

$$M_{i+1} \in M_i + \text{span}\{R_i\} \text{ s.t. } R_{i+1} := I_n - AM_{i+1} \perp A \text{span}\{R_i\}.$$

- ▶ Sparse matrix iterations are conducted using sparse data structures with sparse basic linear algebraic kernels.

As a result, the sparsity pattern evolves naturally from  $M_i$  to  $M_{i+1}$ .

- ▶ Even "learning" sparsity through such sparse matrix iterations can sometimes fail to produce good SPAs.

Chow, E., & Saad, Y. (1998). Approximate inverse preconditioners via sparse-sparse iterations. *SIAM Journal on Scientific Computing*, 19(3), 995-1023.

## Other variants of sparse matrix iteration

- ▶ Venkovic and Anzt (2025) introduce LOMR iterations as a locally enriched variant of MR:

$$M_{i+1} := \arg \min_{M \in M_i + \text{span}\{P_{i-1}, R_i\}} \|I_n - AM\|_F \quad \text{for } i = 0, 1, \dots$$

with (sparse) search directions  $P_0, P_1, \dots$  updated by:

$$M_{i+1} := M_i + \underbrace{\delta_i R_i + \gamma_i P_{i-1}}_{P_i} \quad \text{for } i = 0, 1, \dots$$

- ▶ In that work, we also advocate for the computation of SPAs using sparse matrix conjugate gradient (CG) iterations:

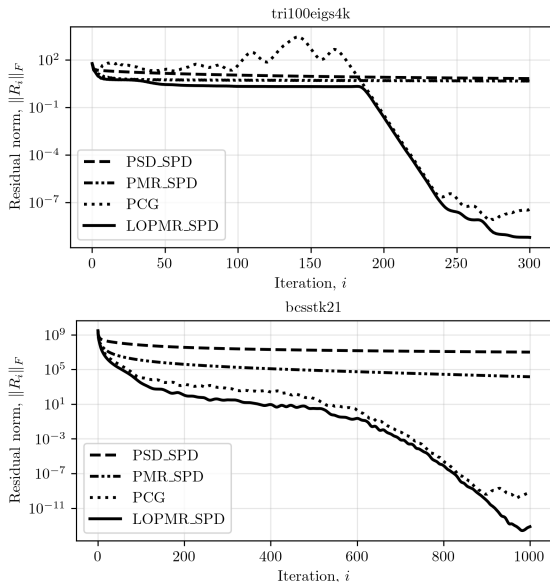
$$M_i \in M_0 + \mathcal{K}_i(A, M_0) \text{ s.t. } R_i := I_n - AM_i \perp \mathcal{K}_i(A, M_0)$$

where  $\mathcal{K}_i(A, M_0) := \text{span}\{M_0, AM_0, \dots, A^{i-1}M_0\}$  is a Krylov subspace.

N. Venkovic and H. Anzt (2025) Global iterative methods for sparse approximate inverses of symmetric positive-definite matrices. arXiv preprint, arXiv:2511.09753.

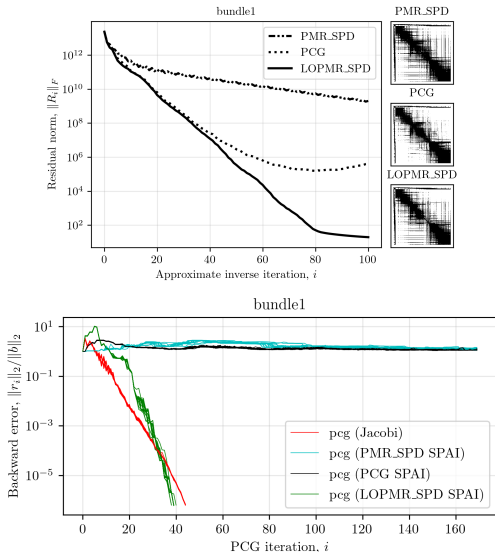
# Comparison of matrix iterations

- LOMR typically acts as a lower envelope to matrix CG iterations:



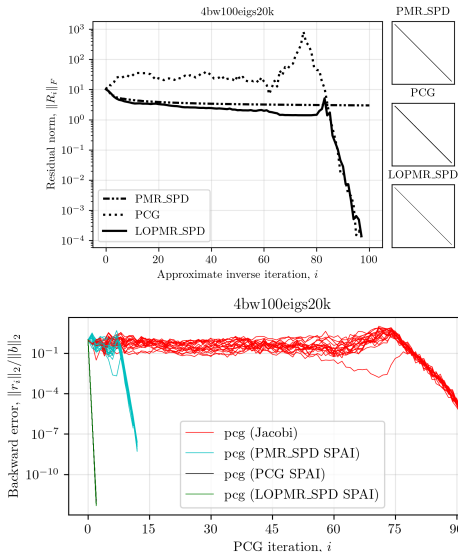
## Results with nonzero dropping

- For some matrices, SPAs obtained by matrix iteration fail to consistently outperform Jacobi preconditioners:



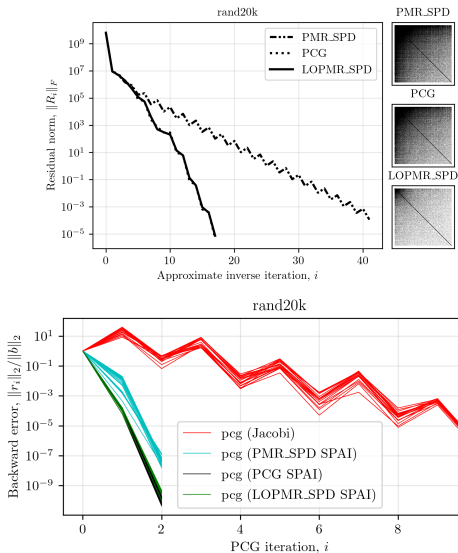
## Results with nonzero dropping, cont'd

- For some other matrices, SPAs obtained by matrix iteration consistently outperform Jacobi preconditioners:



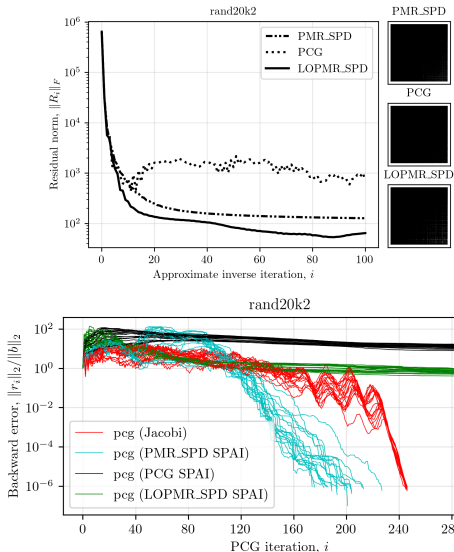
# Results with nonzero dropping, cont'd

- For some matrices, all three matrix iterations produce equally good SPAIs:



## Results with nonzero dropping, cont'd

- For some matrices, the dropping of nonzero components plays a crucial role in finding a good SPAI:



# SparseApproximateInverses.jl

- All these methods are made available on this repository:

venkovic / SparseApproximateInverses.jl Public

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Code

venkovic added compat entry for Julia. 360fe63 · 7 hours ago 64 Commits

|               |                                                    |             |
|---------------|----------------------------------------------------|-------------|
| data/mtcs/spd | added missing spd matrices.                        | last week   |
| examples      | edited matrices grant.                             | yesterday   |
| src           | added missing residual evaluation in pmr_spd_spai. | 4 days ago  |
| LICENSE       | added licence, Manifest and Project files.         | last week   |
| Manifest.toml | added licence, Manifest and Project files.         | last week   |
| Project.toml  | added compat entry for Julia.                      | 7 hours ago |
| README.md     | edited arxiv reference.                            | last week   |

README MIT license

N. Venkovic and H. Anzt (2025) Global iterative methods for sparse approximate inverses of symmetric positive-definite matrices. arXiv preprint, arXiv:2511.09753.

About

Iterative methods to learn and compute sparse approximate matrix inverses (SPAI)s.

Readme

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Activity

2 stars

0 watching

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Report repository

Releases

No releases published

Contributors 1

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Languages

Julia 100%

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## What else is going on?

- The computational cost of every matrix iteration is significant:

| Method | Operation count per iteration              |
|--------|--------------------------------------------|
| MR     | 2 SpGEMM + 1 SpFIP + 1 SpFN + 2 SpGEAMs    |
| SD     | 3 SpGEMMs + 1 SpFIP + 1 SpFN + 2 SpGEAMs   |
| CG     | 2 SpGEMMs + 1 SpFIP + 1 SpFN + 3 SpGEAMs   |
| LOMR   | 3 SpGEMMs + 3 SpFIPs + 2 SpFNs + 4 SpGEAMs |

$\dagger$ : General product of sparse matrices    $\circ$ : Frobenius inner product of sparse matrices

$\bullet$ : Frobenius norm of sparse matrix    $*$ : General addition of sparse matrices and we

are currently investigating ways to alleviate those costs.

### Mixed precision:

- Do as many costly parts of matrix iterations as possible in lower precision arithmetic.
- Coupling the learning of the sparsity pattern in low precision with best-SPAI computation for said pattern works in some cases.

### Self-preconditioned variants:

- Outer-inner iteration parallelizable variant where the computed SPAI is used as preconditioner to improve its quality.

# Approximate low-rank CP tensor representation

## Low-rank CP tensor approximation

- ▶ To construct low-rank CP tensor approximations of the  $d$ -way array  $\mathfrak{X} \in \mathbb{R}^{n_1 \times \dots \times n_d}$ , we make use of the non-convex objective function

$$g : (U_1, \dots, U_d) \mapsto \|\mathfrak{X} - \llbracket U_1, \dots, U_d \rrbracket\|^2$$

and its the gradients

$$\nabla_{U_\ell} g : \mathbb{R}^{n_1 \times r} \times \dots \times \mathbb{R}^{n_d \times r} \rightarrow \mathbb{R}^{n_\ell \times r} \text{ for } \ell = 1, \dots, d.$$

- ▶ A Petrov-Galerkin gradient descent is formulated with approximates

$$U_{\ell,i+1} \in U_{\ell,i} + \text{span} \{ \nabla_{U_\ell} g(U_{1,i}, \dots, U_{d,i}) \}$$

where  $r \cdot n_\ell$  orthogonality constraints need be applied to close the system for  $\ell = 1, \dots, d$ .

For matrices ( $d = 2$ ), we proceed with the following constraints

$$\begin{aligned} R_{i+1} V_i &\perp \text{span} \{ R_i V_i \} \\ R_{i+1}^T U_i &\perp \text{span} \{ R_i^T U_i \} \end{aligned}$$

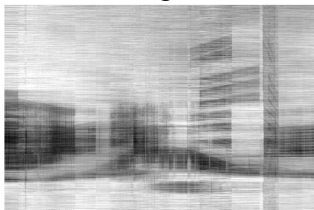
which generalize to  $d$ -way arrays.

## Comparison of methods

- ▶ Similarly as for approximate inverses, the Petrov-Galerkin gradient descent can be made locally optimal for faster convergence.

Sample low-rank approximations achieved in fixed time (5 seconds):

Petrov-Galerkin gradient descent



Locally optimal variant



Currently investigating applications of locally optimal variants for structured low-rank approximation.

E.g., recovering sparse low-rank factors.